

KAPUSKASING, ON, Canada

WMO: 712990

Lat: 49.4053N

Lon: 82.4431W

Elev: 218

StdP: 98.73

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.3	-31.3	-38.4	0.1	-34.1	-35.4	0.1	-31.1	9.6	-12.4	8.9	-12.0	1.6	290	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	29.0	19.7	27.2	18.7	25.5	17.8	21.3	26.6	20.2	25.0	19.2	23.6	5.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.7	23.7	18.5	13.7	22.8	17.4	12.8	21.7	63.2	26.5	59.2	24.9	55.5	23.7	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.3	7.5	-38.7	32.4	3.2	1.9	-41.0	33.8	-42.9	35.0	-44.7	36.1	-47.0	37.5		
			-38.8	23.4	3.2	1.0	-41.1	24.2	-43.0	24.8	-44.8	25.4	-47.1	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-17.4	-15.2	-8.0	0.5	9.1	15.1	17.6	16.3	11.9	4.6	-3.5	-12.0
	DBStd	13.59	8.28	7.47	7.84	5.85	5.28	4.60	3.63	3.78	4.79	4.96	6.39	7.76
	HDD10.0	3806	851	707	561	289	83	9	1	2	33	182	406	683
	HDD18.3	6179	1109	940	818	536	291	117	59	86	200	426	656	941
	CDD10.0	758	0	0	1	3	54	162	236	197	91	16	1	0
	CDD18.3	88	0	0	0	0	4	19	36	23	7	0	0	0
	CDH23.3	921	0	0	1	2	70	257	334	201	55	2	0	0
	CDH26.7	214	0	0	0	0	18	68	78	41	9	0	0	0
Wind	WSAvg	3.6	3.5	3.7	3.7	3.8	3.7	3.3	3.2	3.2	3.4	3.7	3.9	3.6
Precipitation	PrecAvg	850	43	40	56	55	76	90	102	96	95	82	73	52
	PrecMax	1146	119	83	129	132	180	180	211	212	158	188	155	85
	PrecMin	627	9	12	17	11	11	44	24	43	23	27	28	32
	PrecStd	134	21	19	27	29	38	35	41	42	35	36	31	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	5.6	14.9	20.5	29.2	31.6	31.4	30.0	27.9	21.9	13.7	4.5
		MCWB	1.8	3.1	9.3	11.1	18.4	20.1	20.9	20.5	19.6	16.1	10.3	3.0
	2%	DB	0.7	1.9	9.1	16.1	25.0	28.5	28.7	27.6	24.8	18.4	9.7	2.2
		MCWB	0.1	0.3	5.1	8.1	15.3	18.5	19.9	19.6	18.5	13.7	7.4	1.3
	5%	DB	-1.1	-0.7	5.8	12.6	22.1	26.3	26.8	25.6	22.6	15.4	7.0	0.8
		MCWB	-1.8	-2.0	2.5	6.4	13.6	17.0	19.0	18.7	17.4	12.1	5.0	0.1
	10%	DB	-4.2	-3.3	3.1	9.4	19.1	24.0	25.0	23.6	20.1	12.5	4.7	-0.7
		MCWB	-4.8	-4.3	0.7	4.4	12.0	16.2	18.1	17.6	16.0	10.0	3.4	-1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	3.6	10.6	12.3	20.4	22.3	23.0	22.4	21.3	16.9	10.9	3.4
		MCDB	2.6	5.3	13.5	17.0	26.4	29.7	28.0	27.4	25.4	20.4	12.6	4.3
	2%	WB	0.3	0.3	5.2	9.1	16.7	20.3	21.3	20.8	19.4	14.6	7.9	1.4
		MCDB	0.6	1.8	8.3	15.3	22.9	25.9	26.7	25.2	23.2	17.5	9.5	2.1
	5%	WB	-1.8	-1.9	2.7	6.9	14.5	18.7	20.2	19.7	18.1	12.4	5.2	0.2
		MCDB	-1.2	-0.8	5.3	11.8	19.8	23.5	25.0	24.3	21.6	15.0	6.6	0.7
	10%	WB	-4.8	-4.3	0.8	4.9	12.8	17.2	19.1	18.5	16.4	9.9	3.4	-1.3
		MCDB	-4.1	-3.3	3.0	9.0	18.1	22.1	23.3	22.8	19.8	12.6	4.8	-0.7
Mean Daily Temperature Range	5% DB	MDBR	11.2	12.8	13.1	11.9	13.0	12.8	11.9	11.4	10.5	7.9	7.1	8.8
		MCDBR	9.2	11.7	13.9	16.9	18.0	16.2	14.2	14.0	13.4	12.5	9.4	7.9
		MCWBR	9.1	10.4	9.9	10.6	10.0	8.3	7.3	7.4	8.0	8.6	7.6	7.5
	5% WB	MCDBR	9.1	11.2	12.1	14.7	14.9	12.5	11.8	11.8	11.8	11.4	8.6	7.7
		MCWBR	9.1	10.1	9.1	9.8	9.3	7.6	7.1	7.0	7.9	8.5	7.5	7.4
Clear-Sky Solar Irradiance	taub	0.268	0.266	0.299	0.339	0.361	0.372	0.395	0.374	0.365	0.319	0.282	0.254	
	taud	2.251	2.279	2.268	2.327	2.404	2.393	2.352	2.410	2.425	2.535	2.500	2.353	
	Ebn at Noon	782	888	912	908	898	888	858	863	832	824	783	761	
	Edh at Noon	72	89	109	115	112	114	117	106	94	71	58	57	
All-Sky Solar Radiation	RadAvg	1.23	2.32	3.74	4.71	5.18	5.59	5.43	4.54	3.30	1.93	1.03	0.82	
	RadStd	0.18	0.22	0.29	0.38	0.41	0.46	0.54	0.31	0.29	0.25	0.10	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page